

Kunal Singh

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SKILLS

Languages: Python, C++, Java, C, JavaScript, Kotlin

Backend & Databases: FastAPI, Django, Spring Boot, Node.js, MySQL, PostgreSQL, Redis, PostGIS, REST APIs

AI / Data: RAG pipelines, vector embeddings, web scraping (HTTPX, Selenium)

Infra & DevOps: AWS EC2, AWS Secrets Manager, CloudFront, Nginx, OAuth2-Proxy, PM2, Supervisorctl, GitLab EE (source-installed), DevOps

Auth & Cloud: JWT-based RBAC, Google OAuth

Systems: Multithreading, Socket Programming, Performance Profiling (gprof, Google Benchmark)

EDUCATION

Army Institute of Technology (AIT), Pune — BE in Electronics and Telecommunication (E&TC)

CGPA: 9.0/10 | August 2022 – May 2026

Relevant coursework: Data Structures and Algorithms, Object-Oriented Programming, Computer Networks, Database Management Systems, Functional Programming, Control Systems.

ACHIEVEMENTS

- ICPC Asia-West Regional 2023 — Ranked 1st in college, competing against 14,000+ participants.
- Mastercard Code for Change — Finalist among 2,000+ participants.
- Awarded Rs. 75,000 in Merit-cum-Means and Open Merit Scholarships.

EXPERIENCE

Humming Bird Web Solutions — Software Developer L2

April 2026 – Present

- Developed an AI agent platform for website-specific chatbots using web scraping and RAG pipelines with HTTPX, Selenium, FastAPI, and vector embeddings, ingesting data from 35,000+ webpages across 10 chatbot deployments.
- Implemented JWT-based Role-Based Access Control and Google OAuth authentication in FastAPI, enabling separate Admin, Tester, and User access levels for the platform.
- Designed a centralized Configuration Manager integrated with AWS Secrets Manager, replacing local environment variables and enabling automated secret rotation across 10 deployments.
- Deployed and operated the company's internal multi-app platform on AWS EC2 (monolithic architecture) using PM2 and Supervisorctl for frontend/backend process management, avoiding hand-written systemd units, with CloudFront caching and Nginx + OAuth2-Proxy for header-based access verification.
- Extended the company's self-hosted GitLab EE (source-installed, not Omnibus) with a custom worklogs feature — a board showing each user's active issues with inline time logging, story points, and a priority dropdown — working as both feature developer and DevOps engineer on the deployment.

Tarana Wireless — C/C++ Developer

June 2025 – February 2026

- Migrated a production gNMI telemetry pipeline to YANG-based protobuf models, reducing payload size by 49% and eliminating runtime path validation errors through compile-time schema enforcement.
- Reduced function runtime in latency-critical execution paths from 12us to 4us (66%) by optimizing memory allocation and removing redundant vector copies.
- Profiled CPU hotspots with gprof and Google Benchmark, resolving heap allocation bottlenecks to improve telemetry throughput.
- Engineered an RN emulation framework to simulate production network behavior, reducing hardware debugging dependency and accelerating troubleshooting cycles by 35%.

ZF Group — Software Developer Intern

March 2025 – May 2025

- Engineered 12+ REST API endpoints using Java Spring Boot and MySQL to power BOM-based risk dashboards.
- Lowered API latency by 25% by implementing indexed queries and optimizing SQL joins.

Sphyzee — Software Developer Intern

December 2024 – February 2025

- Decreased manual dashboard setup effort by 40% by engineering JSON-driven Power BI pipelines.
- Shortened dashboard migration turnaround time by 30% using automation scripts.

PROJECTS

Scalable Cab Sharing Backend — github.com/kunal763/smart-ride-sharing

August 2024 – November 2024

- Achieved a throughput of 568 requests per second under a load of 10,000 simulated concurrent users during stress testing.
- Cut database read load by 45% by implementing Redis caching for frequently accessed ride IDs.
- Executed geospatial proximity queries using PostGIS to match drivers and riders by radius parameters.